



ĐẠI HỌC BÁCH KHOA HÀ NỘI
VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

Discrete Mathematics

PART 1

COMBINATORIAL THEORY

(Lý thuyết tổ hợp)

PART 2

GRAPH THEORY

(Lý thuyết đồ thị)

Contents of Part 1

Chapter 0: Sets, Relations

Chapter 1: Counting problem

Chapter 2: Existence problem

Chapter 3: Enumeration problem

Chapter 4: Combinatorial optimization problem

Content

- 1. Introduction to existence problems**
2. Basic proof methods
3. Dirichlet principle (pigeonhole principle)

1. Introduction to existence problems

- In the “Counting problem” chapter, we focused on counting the combinatorial configurations. In those problems, the existence of the configurations is obvious, and the main object is to count the number of elements that satisfy the given properties.
- However, in many combinatorial problems, it is very difficult to point out the existence of a configuration that satisfies given properties:
 - For example, when a player needs to calculate his moves to answer whether there is a possibility of winning or not?
 - A person needs to search for the key to decipher a secret code that he does not know if this is really the opponent's encrypted message, or just the secret code issued by the opponent to ensure the safety of real telegrams ...
- In combinatorics, besides the counting problem, there is another very important problem is considering the existence of combinatorial configurations satisfying given properties - the problem of existence.

The 36 officers problem

- This problem is proposed by Euler, it is described as following:

“You're in command of an army that consists of six regiments, each containing six officers of six different ranks. Can you arrange the officers in a 6x6 square so that each row and also each column of the square holds officers of all 6 ranks and all 6 regiments?”

The 36 officers problem

- Using:
 - A, B, C, D, E, F refer to 6 regiments, respectively
 - 1, 2, 3, 4, 5, 6 refer to 6 ranks of officers, respectively.
- This problem could be generalized by replacing 6 by n .
- In the case $n = 4$, a solution to the problem is:

A1	B2	C3	D4
B4	A3	D2	C1
C2	D1	A4	B3
D3	C4	B1	A2

- In the case $n = 5$, a solution to the problem is:

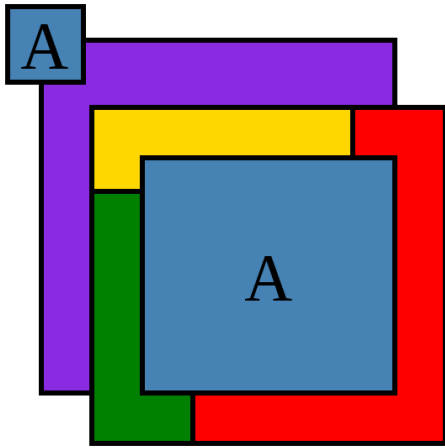
A1	B2	C3	D4	E5
C4	D5	E1	A2	B3
E2	A3	B4	C5	D1
B5	C1	D2	E3	A4
D3	E4	A5	B1	C2

The 36 officers problem

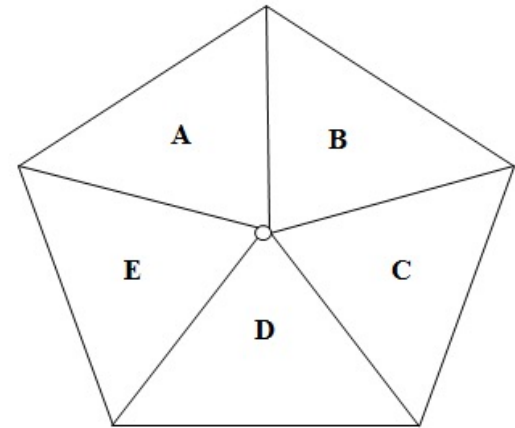
- As the solution to the problem can be represented by combining two same size squares array of symbols: one with upper-case letter and the other with numbers. The problem is also called as Graeco-Latin square (also called an Euler square)
- Euler struggled to find the solution to the 36 officers problem but was unsuccessful. However, the solution exists when $n = 4, 5$ and 7 . Thus, he proposed a general hypothesis: The Graeco-Latin squares are impossible if $n = 4k + 2$.
- Tarri, in 1901, proves the correct hypothesis for $n = 6$, by examining all possible ratings.
- In 1960, three American mathematicians, Boce, Parker, and Srikanda showed a solution with $n = 10$ and then showed a method to construct Graeco-Latin squares for all $n = 4k + 2$, with $k > 1$.

Four color theorem

- There are problems that its content could be explained to anyone, but its solutions anyone can try to find, but it is difficult to find. The 4-color problem is such a problem.
- The problem can be stated visually as follows: Prove that all maps on the plane could be colored with 4 colors so that no two adjacent regions (neighbor regions) are colored by the same color.
- Note that each country is considered a **connected region** and the two countries are called neighbors if they **share a border of a continuous line**.



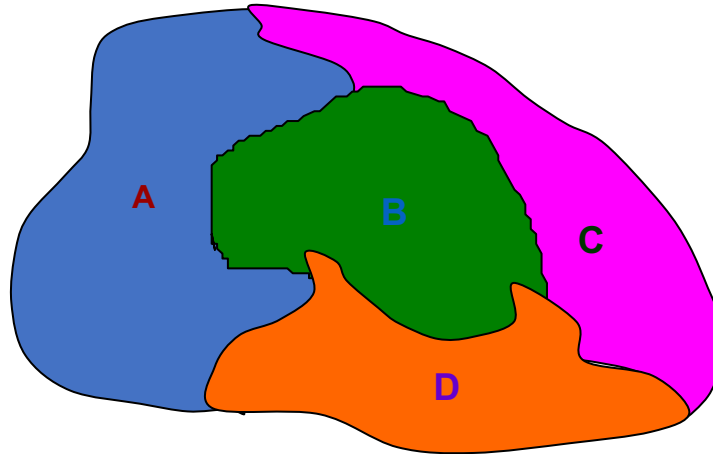
A is not connected region



A and C are not neighbors

Four color theorem

- The number 4 is not random. It has been shown that all maps are colored with color numbers greater than 4, but with color numbers less than 4 it cannot be colored. For example, the map with 4 countries in the picture below cannot be colored if using less than 4 colors.



Four color theorem

- This problem was first proposed in 1852 by Francis Guthrie, while trying to color the map of counties of England.
- In 110 years many proofs have been published but have been flawed.
- In 1976, Appel and Haken gave the proof by computer:

K. Appel and W. Hankin, "Every planar map is 4-colorable," Bulletin of the AMS, Volume 82 (1976), 711-712.

Content

1. Introduction to existence problems
- 2. Basic proof methods**
3. Dirichlet principle (pigeonhole principle)

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction

2.1. Direct proofs

We begin with an example demonstrating the transitivity of divisibility.

Theorem. If a divides b and b divides c then a divides c .

Prove. By using the definition of the divisibility, there exist integers k_1 and k_2 such that

$$a = b k_1 \text{ and } b = c k_2.$$

Then

$$a = b k_2 = c k_1 k_2.$$

Let $k = k_1 k_2$. We have k as an integer, and $a = ck$, so by the definition of divisibility, a divides c .

2.1. Direct proofs

If P, Then Q

- In most theorems, exercises or tests, you need to prove the form "If P, Then Q".
- In this example: “if a divides b and b divides c, then a divides c”
 - "P" is "If a divides b and b divides c" and "Q" is "a divides c".
- This is the standard state of many theorems.
- The direct proof can be conceived as a series of inferences beginning with "P" and ending with "Q":

$$P \Rightarrow \dots \Rightarrow Q$$

Most of proof is direct. When you have to prove, try starting with direct proof, unless you have a good reason not to.

Example 1

- A **rational number** is a number that equals the quotient of two integers.
- Let $\mathbb{Q}$ denote the set of rational numbers.

Theorem: The sum of two rational numbers is rational.

- Proof:
 - Let r and s be rational numbers.
 - Let $r = a/b$ and $s = c/d$, where a, b, c, d are integers, where $b, d > 0$
 - Then $r + s = (ad + bc)/bd$.
 - We know that $ad + bc$ is an integer.
 - We know that bd is an integer.
 - We also know that $bd \neq 0$.
 - Therefore, $r + s$ is a rational number.

Example 1

- A **rational number** is a number that equals the quotient of two integers.
- Let $\mathbb{Q}$ denote the set of rational numbers.

Theorem: Between every two distinct rationals, there is a rational.

- Proof:
 - Let $r, s \in \mathbb{Q}$.
 - Without loss of generality, we may assume $r < s$.
 - Let $t = (r + s)/2$.
 - Then $t \in \mathbb{Q}$. (proof?)
 - We must show that $r < t < s$:
 - Since $r < s$, it follows that
$$2r < r + s < 2s$$
 - Then divide by 2 to get
$$r < (r + s)/2 < s$$
 - Therefore, $r < t < s$.

Example 2

- Prove that if n is an integer and n^3+5 is odd, then n is even

Proof:

- $n^3+5 = 2k+1$ for some integer k (definition of odd numbers)
- $n^3 = 2k-4$
- $n = \sqrt[3]{2k-4}$

Umm, so direct proof didn't work out.

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction

2.2. Proof by Contradiction

Requirement: Proof the statement P

Proof by contradiction:

- Assume it is false (Assume $\neg P$)
- Prove that $\neg P$ cannot occur
 - (it means a contradiction exists: not satisfying the properties given in the problem or come to the absurd such as $1 = 0$)

Requirement: Proof “If P, Then Q”,

Proof by contradiction:

- Assume it is false (Assume that "P and Not Q" are true).
- It thus means a contradiction exists.

Example 1

- Prove that if n is an integer and n^3+5 is odd, then n is even

Proof:

- $n^3+5 = 2k+1$ for some integer k (definition of odd numbers)
- $n^3 = 2k-4$
- $n = \sqrt[3]{2k-4}$

Umm, so direct proof didn't work out.

Rephrased: If n^3+5 is odd, then n is even

Proof by contradiction:

Assume that n^3+5 is odd, and n is odd are true

- n is odd $\rightarrow n=2k+1$ for some integer k (definition of odd numbers)
- $n^3+5 = (2k+1)^3+5 = 8k^3+12k^2+6k+6 = 2(4k^3+6k^2+3k+3)$ is odd ?!?!?
 - As $2(4k^3+6k^2+3k+3)$ is 2 times an integer, it must be even
 - Contradiction!

2.2. Proof by Contradiction

Example 2. Given 7 segments with length greater than 10 and less than 100. Prove that one can always find 3 segments that can be assembled into a triangle.

Solution:

Note that, the necessary and sufficient condition for 3 segments to be assembled into a triangle is: the sum of the lengths of two smaller segments must be greater than the length of the largest segment.

Ordering the given 7 segments in ascending order of length, we have:

$$10 < a_1 \leq a_2 \leq \dots \leq a_7 < 100.$$

We need to prove that: from the above ordered sequence, one always can find 3 consecutive segments such that the sum of length of the first 2 segments is greater than that of the last.

Proof by contradiction:

Assume this does not happen (not Q: could not find any 3 consecutive segments such that the sum of length of the first 2 segments is greater than that of the last).

2.2. Proof by Contradiction

- Proof by contradiction: Assume this does not happen (could not find any 3 consecutive segments such that the sum of length of the first 2 segments is greater than that of the last). Thus, we have equalities:

$$a_1 + a_2 \leq a_3,$$

$$a_2 + a_3 \leq a_4,$$

$$a_3 + a_4 \leq a_5,$$

$$a_4 + a_5 \leq a_6,$$

$$a_5 + a_6 \leq a_7.$$

- As a_1, a_2 are greater than 10, we have $a_3 > 20$. As $a_2 > 10$ and $a_3 > 20$, we have $a_4 > 30$, ..., continuing we have $a_5 > 50$, $a_6 > 80$ and $a_7 > 130$.
- The last equalities $a_7 > 130$ is contradiction with the given condition of the problem: all segments has the length less than 100. Thus, the assumption is false.

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction

2.3. Proof by Contrapositive (Chứng minh bằng phản đề)

Proof by contrapositive uses the logical equivalence of two statements “If P then Q” ($P \Rightarrow Q$) and “If not Q then not P” ($\neg Q \Rightarrow \neg P$):

$$(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$$

Example:

The statement “If this is my car, then its color is red”
is equivalent to

“If its color is not red, then it is not my car”.

- Thus, to prove “If P, then Q” by using contrapositive proof, we prove “If not Q then not P”).

Example

Example 1. If x and y are two integer such that $x+y$ is even, then x and y are either both even or odd.

Proof. The contrapositive is

“If x and y are neither both even nor odd, then its sum is odd.”

- As x and y are neither both even nor odd. Without loss of generality, we may assume x is even and y is odd. Then we could find integers k and m such that $x = 2k$ and $y = 2m+1$. Now we calculate $x+y = 2k + 2m + 1 = 2(k+m) + 1$, thus $x+y$ is odd.
- Therefore, we have proven the result by contraposition

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction

2.4. Proof by Mathematical Induction

- This is a very useful proof technique when we have to prove that the proposition $P(n)$ is true for all natural numbers $n \geq n_0$.
- Similar to the "domino effect" principle.

Outline of proof by Induction:

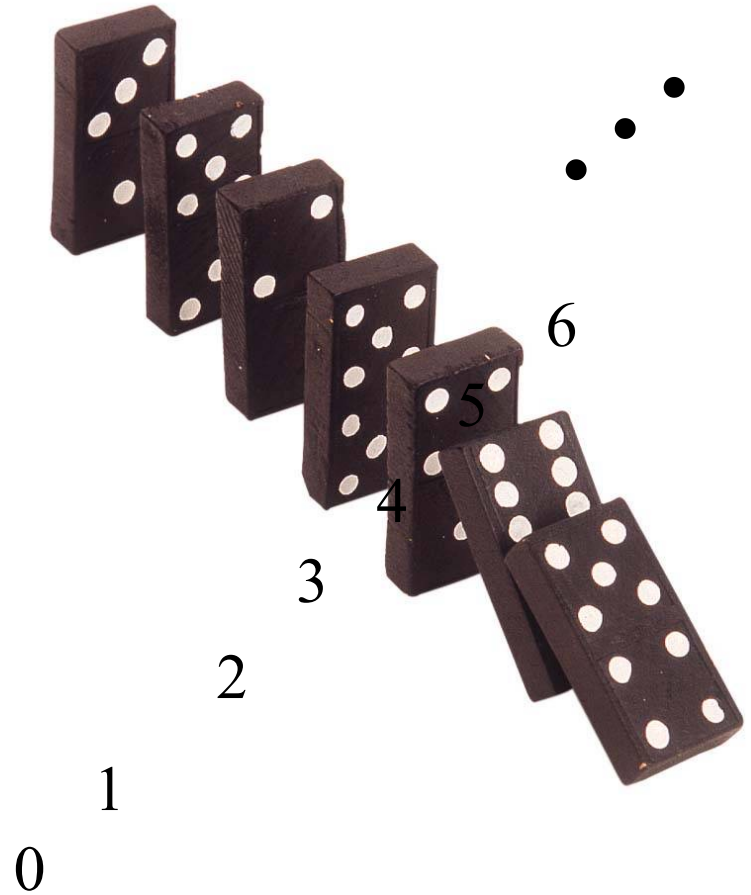
- Basic step: Prove the first statement $P(n_0)$ is true
 - Inductive step: Given any integer $n \geq n_0$, prove that $P(n) \rightarrow P(n+1)$
(Assuming $P(n)$ is true and showing it forces $P(n+1)$ is true)
-
- Conclusion: $P(n)$ is true $\forall n \geq n_0$

(The assumption that $P(n)$ is true is called the **inductive hypothesis**)

The “Domino Effect”

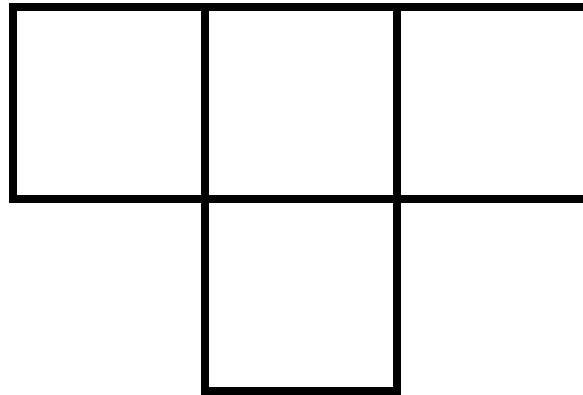
- **Step #1:** Domino #0 falls.
- **Step #2:** For all $n \in \mathbf{N}$, if domino # n falls, then domino # $n+1$ also falls.
- **Conclusion:** All dominos must fall!

Note: This happens even when there are infinitely many dominoes!



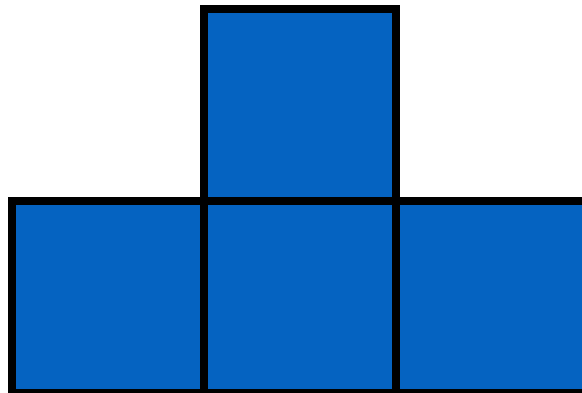
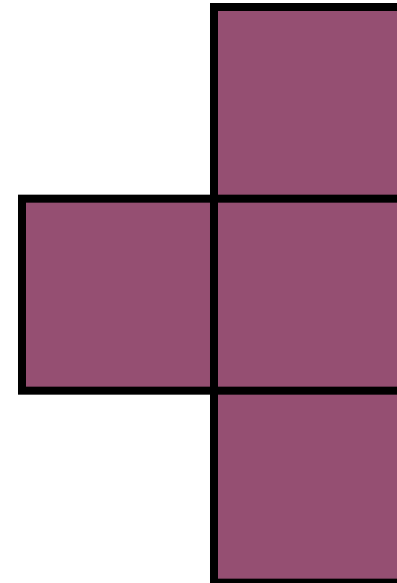
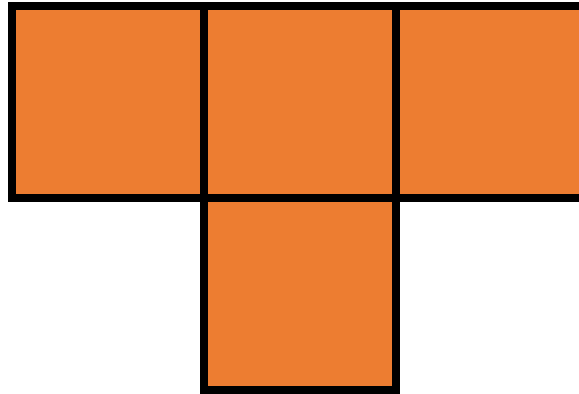
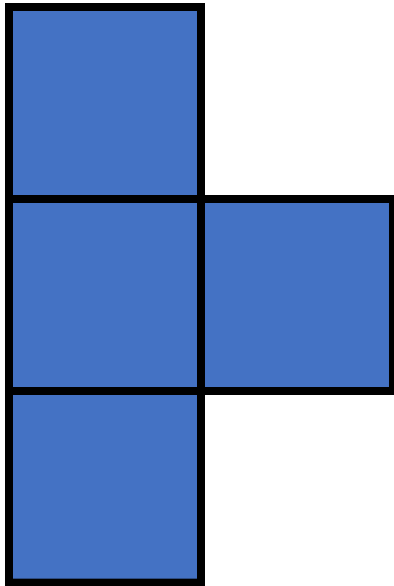
Example

Prove that it is always possible to cover a chessboard of size $2^n \times 2^n$ ($n > 1$) with T-omino cards.

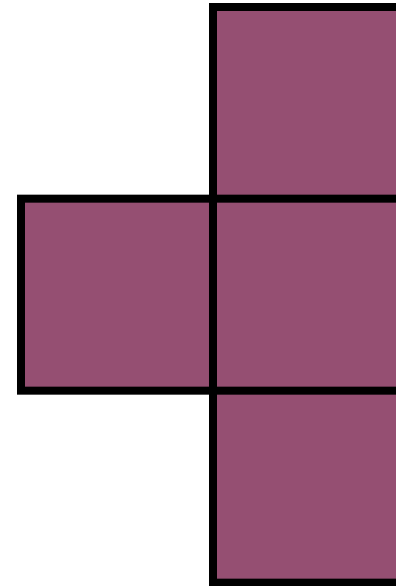
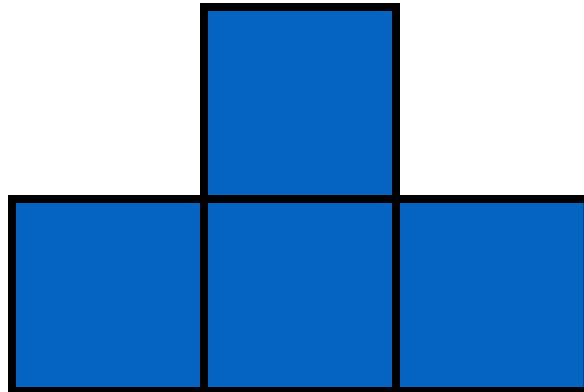
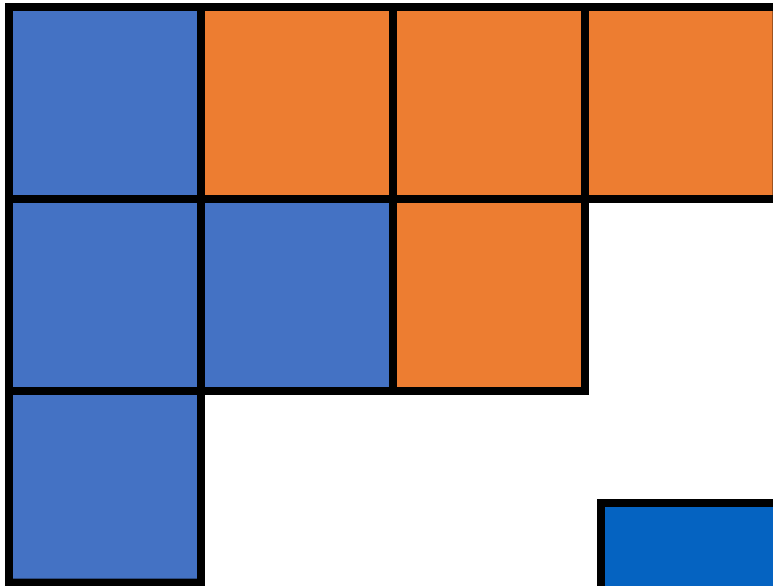


- Basic step: Prove that it is true when $n = 2$

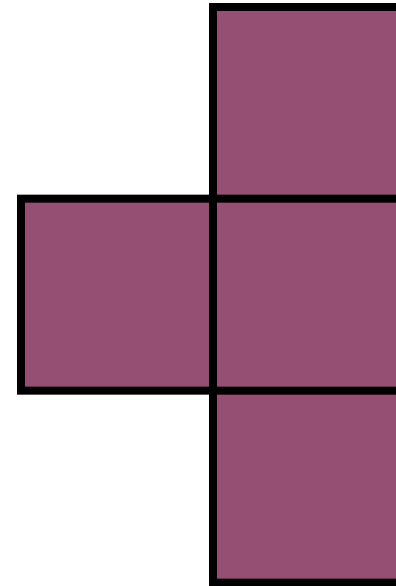
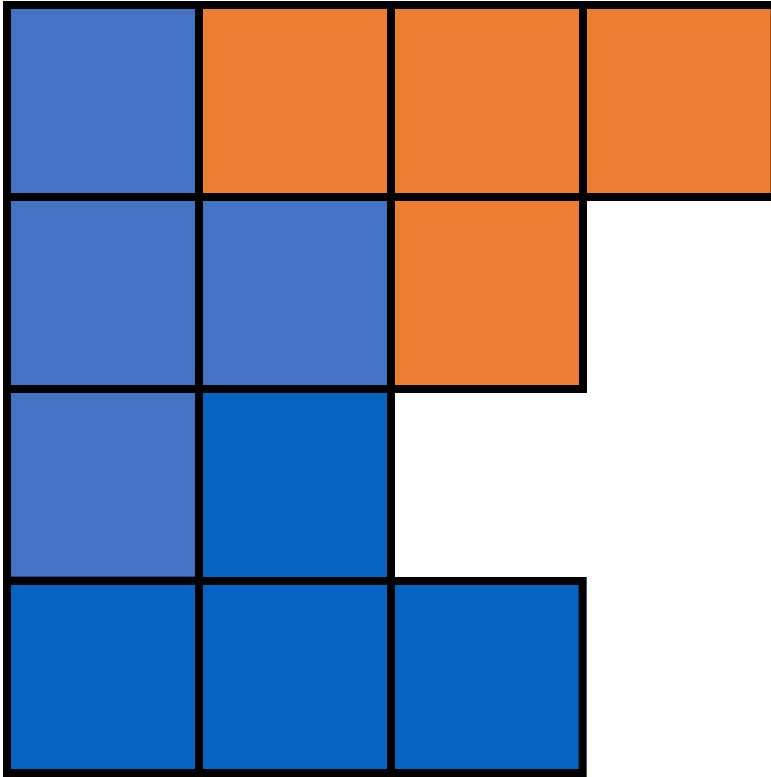
Basic step: Chessboard $2^2 \times 2^2$



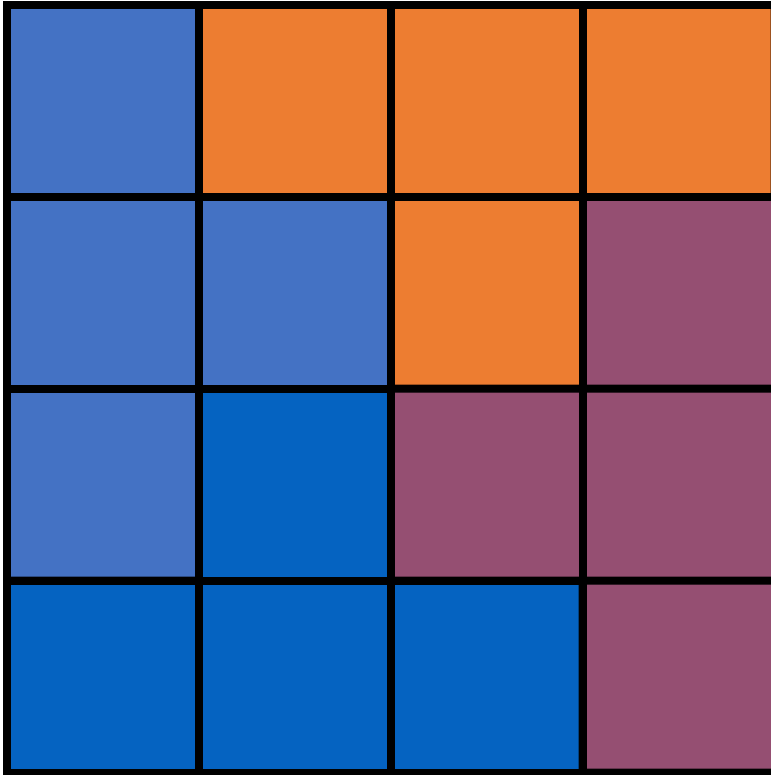
Basic step: Chessboard $2^2 \times 2^2$



Basic step: Chessboard $2^2 \times 2^2$



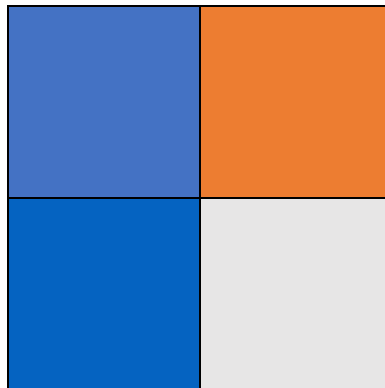
Basic step: Chessboard $2^2 \times 2^2$



Inductive step

Assume we can cover chessboard of size $2^n \times 2^n$. We need to prove also can cover chessboard of size $2^{n+1} \times 2^{n+1}$.

Actually, divide the chessboard of size $2^{n+1} \times 2^{n+1}$ into 4 parts, each of size $2^n \times 2^n$. According to the **inductive hypothesis** (giả thiết qui nạp), each of these parts could be covered by T-ominos. Putting them on the chessboard of size $2^{n+1} \times 2^{n+1}$, we then get the answer for the problem.



Content

1. Introduction to existence problems
2. Basic proof methods
- 3. Dirichlet principle (pigeonhole principle)**

3. Dirichlet principle

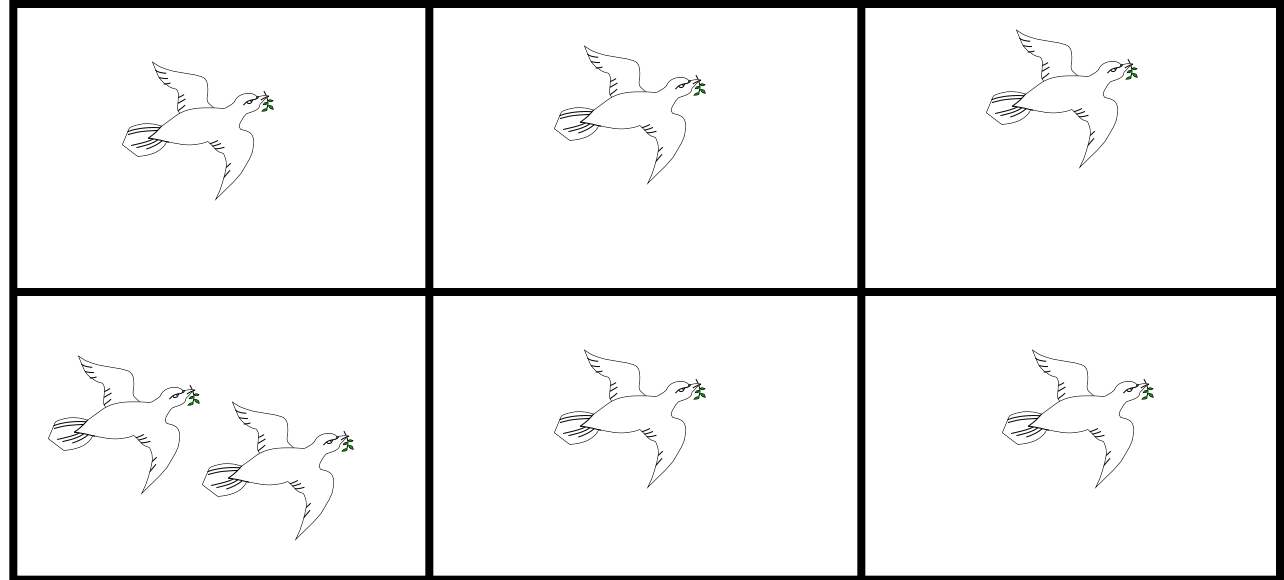
3.1. Principle statement

3.2. Application examples

3.1. Dirichlet principle

If putting more than n objects into n boxes then at least one box has at least 2 objects (≥ 2).

- 7 objects
- 6 boxes



Proof. (Contradiction).

3.1. Dirichlet principle

The above principle has been successfully applied by the German mathematician Dirichlet to solving many existence problems in combinatorics.

It is also presented in the language of pigeons:

“If one put more than n pigeons into n pigeonholes, then at least one hole has more than one pigeon (≥ 2).”

So the principle is also known as " **Pigeonhole principle** ".

Example

If putting more than n objects into n boxes then at least one box has at least 2 objects (≥ 2).

Example 1. Among 13 people, there are always 2 people born in the same month as there are only 12 months.

Example 2. In the exam, the test score is assessed by an integer between 0 and 100. Then at least how many students must take the test so that it is certainly to exist 2 students get same result ?

Solution. There are 101 different results

➔ Using Dirichlet principle, the number of students is 102

Generalized Pigeonhole Principle

Dirichlet principle: If putting more than n objects into n boxes then at least one box has ≥ 2 objects.

When the number of objects putting into k boxes is much larger than the k , it is obviously that the claim in the principle about the existence of a box containing at least two objects is too small. In such a case, we use the following generalized Dirichlet principle:

"If putting n objects into k boxes, one could always find at least one box containing $\geq \lceil n/k \rceil$ objects".

Here the symbol $\lceil \alpha \rceil$ is the least integer greater than or equal to α .

e.g.: $\lceil 3.14 \rceil = 4$

Example

Dirichlet principle: "If putting n objects into k boxes, one could always find at least one box containing $\geq \lceil n/k \rceil$ objects".

Example 3. In a group of 100 people, what is the minimum number of people that were born in the same month ?

Solution: Putting people born in the same month into one group. There are 12 months. Therefore, according to the Dirichlet principle, there exists at least one group consisting $\geq \lceil 100/12 \rceil = 9$ people

Example 4. There are 5 different types of scholarships. What is the minimum number of students so that there are at least 6 students getting the same type of scholarship?

Solution. The minimum number of students needed to ensure that 6 students getting the same scholarship is the smallest integer n such that $\lceil n/5 \rceil = 6$. The smallest integer is $n = 5 \times 5 + 1 = 26$.

Thus, 26 is the smallest number of student needed to ensure that there are 6 students getting the same type of scholarship

Example

Example 5. In a meeting room, there are always 2 people whom are acquainted with the same number of people present at the meeting.

Solution. Let the number of people in the meeting be n , then the number of acquaintance of any people in the meeting could only be 0 to $n-1$.

Obviously, in the room, it could not exists a people having number of acquaintances is 0 (means do not know anybody) and a people having number of acquaintances is $n-1$ (means know everybody in the room). Therefore, dividing people based on the numeber of acquaintances, one could only divide n people into $n-1$ groups.

According to Dirichlet principle, there are at least one group containing at least 2 people, it means one could always find at least 2 people with the same number of acquaintances



Exercises

1. CMR trong số 10 người bất kỳ luôn tìm được hoặc là hai người có tổng số tuổi chia hết cho 16, hoặc là hai người mà hiệu số tuổi của họ chia hết cho 16.
2. Một trung tâm máy tính có 151 máy vi tính. Các máy của trung tâm được đánh số từ 1 đến 300 sao cho không có hai máy nào được đánh cùng số. CMR luôn tìm được hai máy được đánh số bởi các số nguyên liên tiếp.
3. Mỗi dãy gồm n^2+1 số *phân biệt* (nghĩa là các phân tử là khác nhau từng đôi) luôn chứa hoặc dãy con tăng ngặt độ dài $n+1$ hoặc dãy con giảm ngặt độ dài $n+1$.